MILITARY SPECIFICATION

TRANSMITTER, RATE GYROSCOPE TRU-2A/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of remote rate-of-turn gyroscope transmitter, designated TRU-2A/A.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-5541	Chemical Conversion Coatings On Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-I-27619	Indicator, Attitude Director

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:10Z
Check the source to verify that this is the current version before use.

MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipmen
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained for the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.2 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided. Components contained within hermetically sealed containers shall be considered to be adequately protected against deterioration.

3.3.3 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met during storage or normal service.

* 3.3.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4.1.1 Corrosive fumes. The materials, as installed in the transmitter and under the service conditions specified herein, shall not liberate deleterious fumes.

* 3.4 Design and construction. The transmitter shall be designed to sense the rate-of-turn of an aircraft by an output signal that is proportional to the rate at which the aircraft is turning. The transmitter shall be so designed and constructed that the modules or major components may be removed and replaced without the use of special tools unless they are approved by the procuring activity.

3.4.1 The transmitter shall be sufficiently rugged to meet the requirements specified herein and shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2 The design of the transmitter shall not incorporate rubbing parts of components which would be subject to wear (except for bearings).

3.4.3 Shock mount. Shock mounts shall not be used between the transmitter and the airframe.

3.4.4 The transmitter shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases without the use of special tools unless they are approved by the procuring activity.

* 3.4.5 Maintainability. The transmitter shall be easily maintainable. The following maintainability requirements shall be considered in the design of the transmitter:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performanc

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

c. Rapid and positive identification of the replaceable defective part, assembly or component

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

* 3.4.6 Reliability. The transmitter shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.4.6.1 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 3,000 hours before the equipment consistently fails to meet the specified MTBF index.

* 3.4.7 Safety. Hazards associated with use of this equipment shall be controlled as specified in MIL-STD-882.

* 3.5 Performance. The transmitter shall be capable of satisfactory operation when subjected to any combination of the following conditions:

a. Temperatures - Ambient operating temperatures ranging from -65° to +71°C (-85° to +160°F) and operation for 10 minutes at +98°C (+208°F) and storage temperatures ranging from -65° to +85°C (-85° to +185°F)

b. Humidity - Relative humidity up to 100 percent

c. Vibration - Vibration at a double amplitude of 0.060 inch through a frequency range of 10 to 55 cps and at accelerations up to 5g through a frequency range of 5 to 500 cps

d. Acceleration - Acceleration forces up to 20g

e. Altitude - Pressures ranging from 30 inches Hg (sea level) down to 0.082 inch Hg (approximately 80,000 feet)

f. Fungus - Exposure to fungus enzymes

g. Salt-Fog - Exposure to salt fog for a minimum period of 168 hours

h. Dust - Exposure to blowing fine sand and dust particles

i. Temperature shock - Temperature shock caused by abrupt temperature drops from +85° to -40°C (+185° to -40°F)

j. Dynamic shock - Dynamic shocks at accelerations up to 20g for a nominal duration of 10 milliseconds

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

* k. Handling shock - Shocks characteristic of rough handling during bench checkout.

1. Exposure to acoustic sound fields that are characteristic of aircraft.

* 3.5.1 Performance characteristics. The transmitter shall provide electrical signals directly proportional to the rate of yaw of aircraft. The design of the transmitter shall be such that when it is inclined with its forward end elevated, an anticipatory signal, which is a function of rate of roll of the aircraft, will be provided as an output signal while the aircraft is being rolled prior to turning. The inclination of the transmitter at an angle up to 15° shall be accomplished at the time the transmitter is installed in the aircraft. The transmitter shall meet all the performance requirements specified herein when mounted upside down.

3.5.1.1 Compatibility. The transmitter shall be compatible with and shall meet the applicable requirements of MIL-I-27193 or MIL-I-27619, or both.

3.5.2 Radio noise interference and susceptibility. The transmitter shall meet the radio noise interference and susceptibility requirements of MIL-I-6181.

3.5.3 Friction and hysteresis. The friction and hysteresis in the yaw axis shall not exceed .50 percent of the maximum sensitive rate.

* 3.5.4 Signal source impedance. The signal source impedance shall be 500 ohms ±10 percent.

3.5.5 Warmup. The transmitter shall not require a warmup period of more than 90 seconds under any temperature conditions specified herein.

3.5.6 Transient response. Response of the transmitter to a step input shall exhibit a time constant of 1.75 ±0.25 seconds.

3.5.7 Resolution and threshold. Both the resolution and the threshold of the transmitter shall be less than 0.015° per second.

3.5.8 Output signal. The current output into two 1,000-ohm meter movements in parallel shall be 1.0ma ±6.50 percent when the transmitter is rotated in yaw at the rate of 90° per minute. For all other rates-of-turn up to 360° per minute, the current output shall be proportional to the values stated above and the tolerance shall be ±6.50 percent in every case. When the transmitter is rotated to the right in the same manner that an aircraft with the transmitter properly installed would be turned to the right, the polarity of pin C shall be positive. With the precession axis of the transmitter inclined up to 15° with the forward end of the transmitter elevated and the transmitter then rotated in yaw in a horizontal plane, the output signals from the rate

Source: https://assist.dla.mil -- Downloaded: 202640-20T11:10Z
Check the source to verify that this is the current version before use.

transmitter shall be within 3.50 percent of the value obtained when the precession axis of the rate transmitter is parallel to the horizontal plane and the transmitter is rotated in yaw in the horizontal plane.

* 3.5.9 Cross coupling. When rotated in the spin axis at maximum rate, the rate transmitter shall show less than 1-1/2 percent of that value produced when rotated at the same rate in the yaw axis.

3.5.10 Dielectric strength. The complete electrical system shall be so insulated from the transmitter case that it will withstand 500V for a period of 10 seconds.

* 3.5.11 Extreme rate-of-turn. The output of the transmitter shall be limited to a value equivalent to 390° ±30°. Higher rates-of-turn shall not cause damage to the transmitter.

3.5.12 Natural frequency. The undamped natural resonant frequency of the rate sensor shall be more than 5 cps.

3.5.13 Friction and hysteresis. Friction and hysteresis variation at zero input and at maximum transmitter rate-of-turn shall not exceed 20 μa.

3.5.14 Thermal shock. The transmitter case shall show no evidence of leakage when subjected to alternate immersions in water at a temperature of 85°C (185°F) and then in water at a temperature of 5°C (41°F).

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Damping. The transmitter shall be of a symmetrical nongaseous fluid-damped design and shall be temperature compensated without the use of heaters.

3.8 Rate-of-turn signal pickoff. The pickoff shall be so designed that no rubbing of contacting surfaces will be used. It shall be an E- or Microsyn-type pickoff, or equivalent.

3.9 Spin motor. The spin motor shall not incorporate slip rings, contacts, brushes or any similar element that may fail due to wear incurred in normal operation.

* 3.10 Case. The transmitter case shall be made of sturdy material which shall be uniform in texture and shall be finished in durable lusterless black, Color No. 37038 of FED-STD-595. It shall be so rugged that it will not become distorted to such extent as to degrade performance of the transmitter. The case

Source: https://assist.dla.mil -- Downloaded: 202640-20T11:10Z
Check the source to verify that this is the currentversion before use.

shall be marked to indicate the location of the sensitive axis and the spin axis for proper installation of aircraft. The word FRAGILE shall be marked on the top of the case in letters not less than .25 inch high. The gyro portion of the transmitter shall be hermetically sealed and of such design that its internal mechanism may be removed from the case as a unit, replaced, and the case replaced without the use of special tools unless they are approved by the procuring activity. The seal shall be accomplished in such a manner that it will not be dependent upon materials that might be affected by the action of any atmosphere to which the transmitter may be subjected.

* 3.10.1 Filling medium. The filling medium in the case shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) (-85°F) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium or 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. If the gyro is a floated unit and is sealed in a compartment of the case separate from other components, this requirement shall apply only to the compartment containing the gyro wheel.

3.11 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.12 Weight. The total weight of the completely assembled transmitter, less mounting screws and cables, shall not exceed 2.50 pounds.

3.13 Electrical circuit

3.13.1 Power requirements. The transmitter shall be designed to operate from a 28V dc power source as defined in MIL-STD-704.

3.13.2 Power consumption. The peak power consumption of the transmitter shall not exceed 15w. After 90-second warmup period, the power consumption shall not exceed 9w.

3.13.3 Receptacles. The receptacles provided shall be of the Bendix Scintilla PT series, or equal, and shall mate with Scintilla type PT06P-8-4S, or equal.

3.13.4 Pin utilization. The pins of the electrical receptacle shall be wired as follows:

PIN	USE
A	Power ground
B	+28V dc
C	(+) Output signal (transmitter
D	(-) In right turn)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

3.13.5 Electrical transients. The transmitter shall be designed to withstand a spike of ±200V applied between pins A and B for 1 millisecond.

3.13.6 Power inversion. The 28V dc input shall be converted to a regulated ac voltage by means of a static inverter within the instrument case.

* 3.13.7 Undervoltage. The transmitter shall not be damaged by voltage below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.14 Finishes and protective coatings

3.14.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.14.1.1 Small holes and case inserts need not be anodized.

3.14.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.14.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 shall be used in lieu of anodizing.

3.14.1.4 Casting containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.14.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

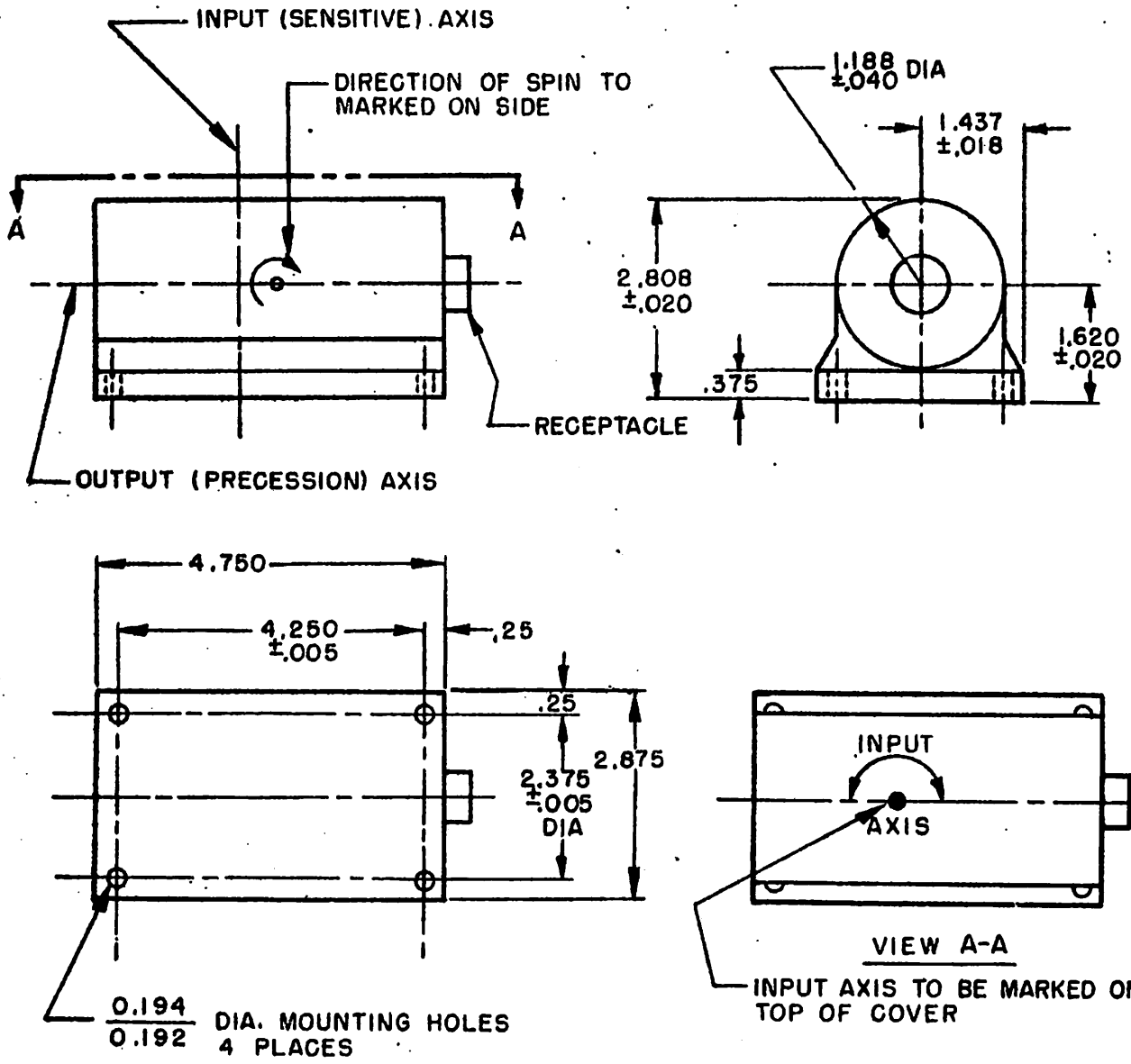
3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The direction of spin of the gyro rotor shall be clearly marked on the case.

3.16 Workmanship. The transmitter, including all parts and accessories, shall be constructed and furnished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and free of parts, burrs and sharp edges as well as resin flash that may crumble.

3.16.1 Dimensions and tolerances. The dimensions and tolerances of the transmitter shall be in accordance with figure 1.

3.16.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES. TOLERANCES: DECIMALS ±.015.

FIGURE 1. Outline and Dimensions

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

3.16.3 Cleaning. The transmitter shall be thoroughly cleansed and loose, spattered, or excess solder, metal chips, and other foreign materials removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C) (77°F). When tests are made with the atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Voltage input. Unless otherwise specified, the transmitter shall be tested with 28V dc input on pins A and B as specified in 3.13.5.

4.3.3 Attitude. Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.3.4 Meter movement for testing. Unless otherwise specified, the transmitter shall be tested while connected to a 500-ohm, ±3 percent, load simulating the meter movements specified in 3.5.8.

4.3.5 Test sequence. Unless otherwise specified, the tests shall be conducted in the sequence listed in 4.6.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T11:10Z
Check the source to verify that this is the current version before use.

4.4 Qualification testing

* 4.4.1 Test samples. The test samples shall consist of 10 transmitters representative of the production equipment. Three of the transmitters shall be subjected to all the tests except reliability and the other seven transmitters shall be subjected to the reliability test only.

* 4.4.1.1 Reliability demonstration. Seven transmitters representative of the production equipment shall be tested as specified in 4.6.28. Accept-reject criteria shall be in accordance with Test Plan 1 of MIL-STD-781 as modified by figure 2.

4.4.2 Test report. When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report - Three copies of a test report in accordance with MIL-STD-831

b. The samples that were tested in the (as is) condition

c. Two sets of assembly drawings

d. Brief operating instructions.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests described under 4.6.

* 4.5 Quality conformance tests. Quality conformance test shall consist of:

a. Individual tests

b. Sampling plans and tests.

* 4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Power consumption

c. Null limits

d. Sensitivity

e. Transient response

11

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

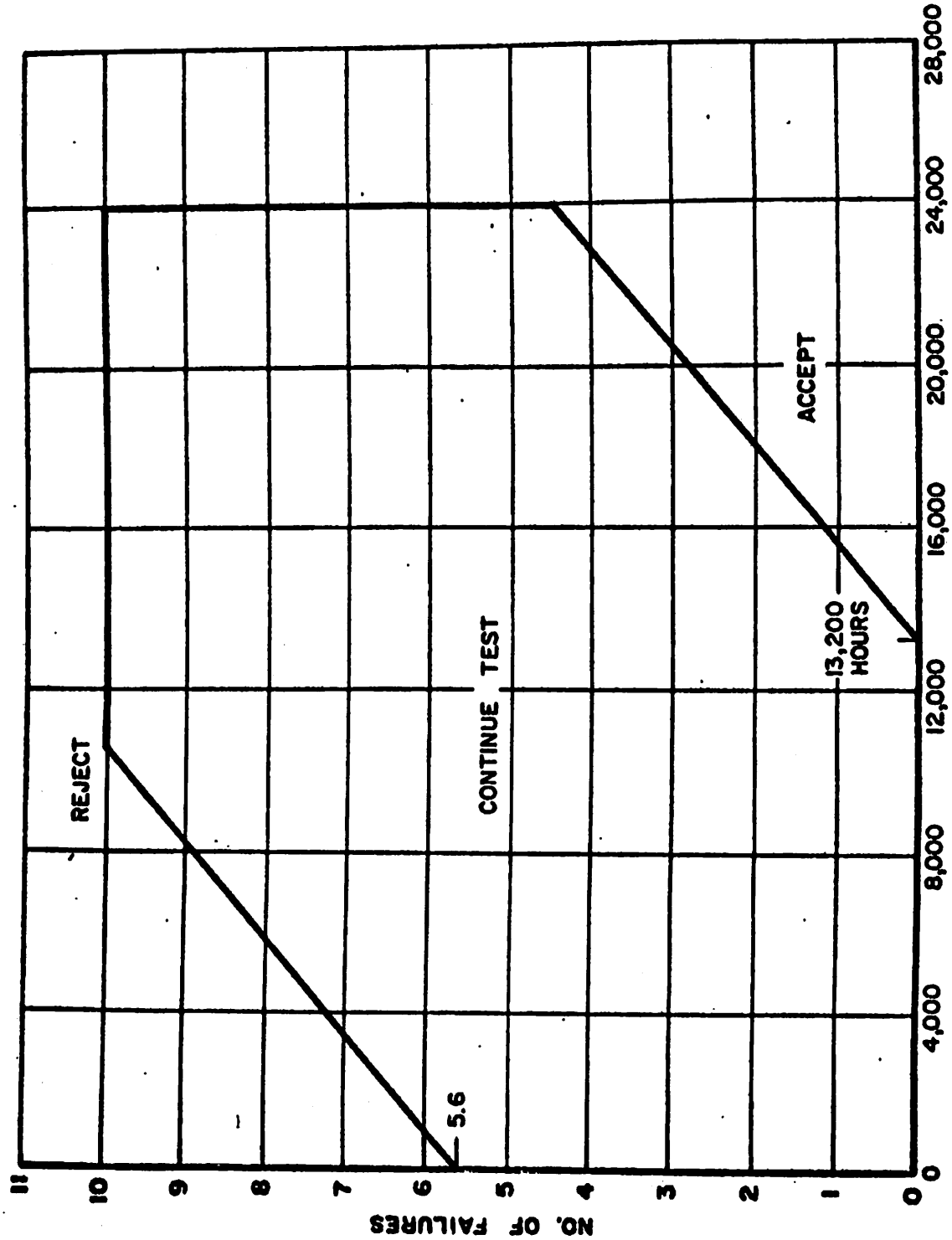


FIGURE 2. Accept Continue-Test Reject Criteria

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

f. Cross coupling

g. Dielectric

h. Resolution and threshold

i. Friction and hystersis

j. Sealing.

4.5.2 Sampling plans and tests

* 4.5.2.1 Sampling plan A. One transmitter shall be selected at random from each 100 or less produced on the contract or order and subject to the following tests as described under 4.6:

a. Individual tests

b. Sensitivity at low temperature

c. Sensitivity at high temperature

d. Vibration error

e. Yaw oscillation

f. Electrical transients.

* 4.5.2.2 Sampling plan B. Unless otherwise specified, three transmitters shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference and susceptibility

c. High temperature

d. High altitude-low temperature

e. Temperature shock

f. Dynamic shock

g. Vibration failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

h. Acceleration

i. Humidity

j. Fungus

k. Salt fog

l. Dust

m. Thermal shock

n. Compatibility

o. Acoustical noise

p. Voltage extremes

q. Output impedance

r. Weight and size.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of defects likely to be found and methods of correcting them.

* 4.5.4 Sampling plan C. Unless otherwise specified, seven transmitters shall be selected from the first 15 on the contract or order. These transmitters shall be tested as specified in 4.6.28 (does not include 4.6.28.1). Accept-reject criteria shall be in accordance with *Test Plan IV of* MIL-STD-781 and shall demonstrate a minimum acceptable MTBF of 2,000 hours.

4.5.4.1 Recording. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

4.6 Test methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and configuration as shown on figure 1.

4.6.2 Power consumption. The transmitter shall be connected for normal operation without power applied and with the gyro rotor at rest. Power shall be applied and the starting power, measured within 5 seconds from the time power is applied, shall not exceed 15w. Ninety seconds subsequent to application of power, the power consumption shall not exceed 9w.

4.6.3 Null limits. When the output signal is fed through a 500 ohm ±3 percent load, the output current with 0° per minute rate of turn shall be zero within ±40 μa.

* 4.6.4 Sensitivity. The output of the rate sensor shall be as follows for inputs about the yaw axis. These readings shall be taken with the output signal fed into a 500-ohm resistance ±3 percent.

Rate-of-Turn (Degrees per minutes)	Direction of Turn	Output Milliamperes
45	cw and ccw	0.5 ±.045
90	cw and ccw	1.0 ±.065
180	cw and ccw	2.0 ±.13
360	cw and ccw	4.0 ±.26

4.6.5 Transient response. While the rate transmitter is rotating at 360° per minute in either direction and the output is as specified above, the turning shall be stopped suddenly. The time for the output current to drop to 1.47 ma shall be measured and shall be within 1.75 to ±0.25 seconds. The test shall be repeated in the opposite direction of rotation.

* 4.6.6 Cross coupling. The rate transmitter shall be rotated in each direction about the spin axis at a rate of 360° per minute. The spin axis shall be inverted by 180° and this test repeated. The total output reading shall not exceed 60 μa.

* 4.6.7 Dielectric strength. The completely assembled transmitter, with radio noise filters installed, shall be subjected to a potential of 500V dc between the positive terminal and conducting parts of the case and then between the negative terminal and conducting parts of the case, each for a period of 10 seconds. There shall be no breakdown of insulation nor any permanent damage to the instrument.

Source: https://assist.dla.mil -- Downloaded: 202640-20T11:10Z
Check the source to verify that this is the current version before use.

* 4.6.8 Voltage extremes. Voltages of 21 and 29V dc shall be applied to the transmitter. The test of 4.6.3 shall be met for both input voltages except the output shall not exceed ±50 µa. While each voltage is applied, the test specified in 4.6.4 shall be repeated, except that all readings shall be within +10 and -17 percent of the normal output values at 90,180 and 360 degrees per minute. The limit for 45°/minute (cw and ccw) shall be ±85 µa.

4.6.8.1 Electrical transients. The transmitter shall be subjected to an electrical spike of 200V for 1-millisecond duration in both positive and negative polarities between pins A and B. The transmitter shall then meet the tests specified in 4.6.2 and 4.6.4.

4.6.9 Resolution and threshold. Both resolution and threshold shall be tested at a rate of 0.015° per second or less so that the perceptible change in output signal will be at least 6 µa.

4.6.10 Friction and hysteresis. The transmitter shall be tested for friction and hysteresis and the total variation at zero input at the maximum rate-of-turn of 360° per minute for a minimum period of 15 seconds in both clockwise (cw) and counterclockwise (ccw) directions, shall not exceed 20 µa.

* 4.6.11 Sensitivity at low temperature. The transmitter shall be properly connected and placed in a chamber at a temperature of -54° ±2°C (-65° ±4°F) for a period of 4 hours. At the end of the 4 hours and while still at low temperature, 28V dc shall be applied to the transmitter. After power has been applied for 10 minutes, the transmitter shall meet the tests specified in 4.6.4, except the tolerance shall be ±10 percent, and 4.6.5, except the tolerance shall be +30 -25 percent. When at rest and at low temperature, subsequent to rotation, the transmitter shall meet the test specified in 4.6.3, except the tolerance shall be ±70 µa.

* 4.6.12 Sensitivity at high temperature. The properly connected transmitter shall be subjected to a temperature of 71° ±2°C (160° ±4°F) for a period of 4 hours. While still at this temperature, 28V dc shall be applied. After 5 minutes of operation, the transmitter shall be subjected to and shall pass the tests specified in 4.6.4, except the tolerance shall be ±10 percent, and 4.6.5, except that the tolerance shall be 20 percent. While still at high temperature, the transmitter shall pass the test specified in 4.6.3, except the tolerance shall be ±70 ua. The temperature shall be increased to 98°C (208°F) and held for 10 minutes. During this time, the transmitter shall be operating and shall indicate the imposed rate-of-turn within the specified limits.

4.6.13 Vibration error. The vibration error test shall be conducted with power applied. The test shall consist of a frequency survey with vibration applied to the longitudinal, lateral, and vertical axes of the transmitter. The transmitter shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 cps in each of

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

the three mutually perpendicular axes except that the vibration shall be limited to 3g. The duration of the vibration shall be 3 minutes in each axis for one survey of the complete frequency range specified herein or 3 minutes with vibration applied in a circular motion for one survey of the complete frequency range specified herein. During this test, the output signal shall not exceed ±50 μa.

4.6.14 Yaw oscillation. The transmitter, with power applied for not less than 90 seconds, shall be subjected to the following yaw oscillations:

a. 2° double amplitude - 8-second period

b. 4° double amplitude - 8-second period

The average value of the signal integrated over at least 10 full cycles shall not exceed 65 μa.

4.6.14.1 Static balance. The transmitter shall be mounted upside down in the test fixture. It shall then be subjected to the tests specified in 4.6.3 through 4.6.6.

* 4.6.15 Sealing. Any gas filled compartment in the transmitter case shall be tested for leaks by means of a mass-spectrometer-type of helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi. This test may be conducted on the compartments prior to final assembly.

4.6.16 Radio noise interference and susceptibility. The transmitter shall pass the radio noise interference and susceptibility tests specified in MIL-I-6181.

4.6.17 High temperature. The high temperature test shall be conducted in accordance with MIL-STD-810, Method 501 for a period of 24 hours. While still at the high temperature, rated power shall be applied. The transmitter shall operate without sticking. After the transmitter has returned to room temperature, it shall pass the tests specified in 4.6.3 and 4.6.4. There shall be no damage that would affect subsequent operation.

* 4.6.18 Temperature-altitude. The transmitter shall be subjected to a temperature-altitude test in accordance with MIL-STD-810 Method 504, class 2, except that the altitude shall be 80,000 feet in step 5, the stabilized temperature shall be maintained at 85°C (185°F) for 4 hours. In step 7, the operating temperature shall be 71°C (160°F). The transmitter shall operate without sticking. After returning to room temperature and atmospheric pressure, the transmitter shall meet the individual tests. The transmitter shall operate properly throughout this test and there shall be no damage that would affect subsequent operation. The limits of 4.6.11 and 4.6.12 shall apply at the respective extreme temperatures.

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:10Z
Check the source to verify that this is the current version before use.

4.6.19 Temperature shock. The temperature shock test shall be conducted in accordance with MIL-STD-810 Method 503. After the transmitter has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The transmitter shall then be subjected to and shall pass the individual tests.

* 4.6.20 Dynamic shock. The dynamic shock tests shall be conducted in accordance with MIL-STD-810 Method 516, procedures I, III and V. Following the test specified in procedure I and after procedure V, the transmitter shall meet the individual tests.

* 4.6.21 Vibration failure. The vibration failure test shall be conducted in accordance with MIL-STD-810 Method 514, curve C, with power applied. At the end of this test, the rate transmitter shall be subjected to and shall pass the tests specified in 4.6.3 and 4.6.4. No damage to the transmitter shall result from this test.

* 4.6.22 Acceleration. The transmitter, not operating, shall be mounted on a centrifuge in its normal operating position and accelerated, first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The test of 4.6.3 shall be met for both input voltages except the output shall not exceed ±50 μa. The transmitter shall be subjected to an acceleration of 20g's in each of the axes for 1 minute. At the conclusion of this test, the transmitter shall be subjected to and shall meet the individual tests. During test of 4.6.4, an additional input rate of at least 900° per minute shall be applied, cw and ccw. The output shall be 3.75 to 5.00 milliamperes. No damage to the transmitter shall result from this test.

* 4.6.23 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810. If the outer case of the gyroscope is not hermetically sealed, the tests shall be conducted on three functional units. If the outer case of the gyroscope is hermetically sealed, the environmental tests may be conducted on three dummy units. At the completion of each test, the transmitter shall pass the individual tests; if functional. Otherwise the transmitter case shall be inspected for any damage that would impair transmitter operation.

4.6.23.1 Humidity. The humidity test shall be conducted in accordance with MIL-STD-810 Method 507, procedure I.

4.6.23.2 Fungus. The fungus test shall be conducted in accordance with MIL-STD-810 Method 508, procedure I.

4.6.23.3 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810 Method 509, procedure I for a period of not less than 168 hours.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

* 4.6.23.4 Dust. The dust test shall be conducted in accordance with MIL-STD-810 Method 510, procedure I.

* 4.6.24 Thermal shock. The transmitter inner case, with mechanisms removed and sealed in the same manner as the complete transmitter, shall be immersed alternately in tap water maintained at 85° ±5°C (185° ±9°F) and 5° ±4°C (41° ±7°F) for a total of eight cycles. The length of time for each bath immersion shall be 30 minutes with not more than 5 seconds elapsing between bath immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the transmitter inner case shall be subjected to and shall meet the test specified in 4.6.15. If it is a floated unit, the inner case shall be inspected and shall show no leakage of fluid after being subjected to the vacuum as specified in 4.6.15.

* 4.6.25 Compatibility. The transmitter, secured to a rate table, shall be connected to an ARU-2B/A or ARU-11/A attitude director indicator in parallel with a 1,000-ohm ±3 percent load. With power applied and with the transmitter rotating at a rate of 90° per minute, the rate-of-turn needle of the indicator shall deflect one needle width ±.03125 inch. When the transmitter is rotated at a rate of 180° per minute, the rate-of-turn needle of the indicator shall deflect two needle widths ±.03125 inch. After rotation has been stopped, the needle shall return to center within ±.03125 inch with no overshoot. The above tests shall be performed with the transmitter rotating both in the cw and ccw directions.

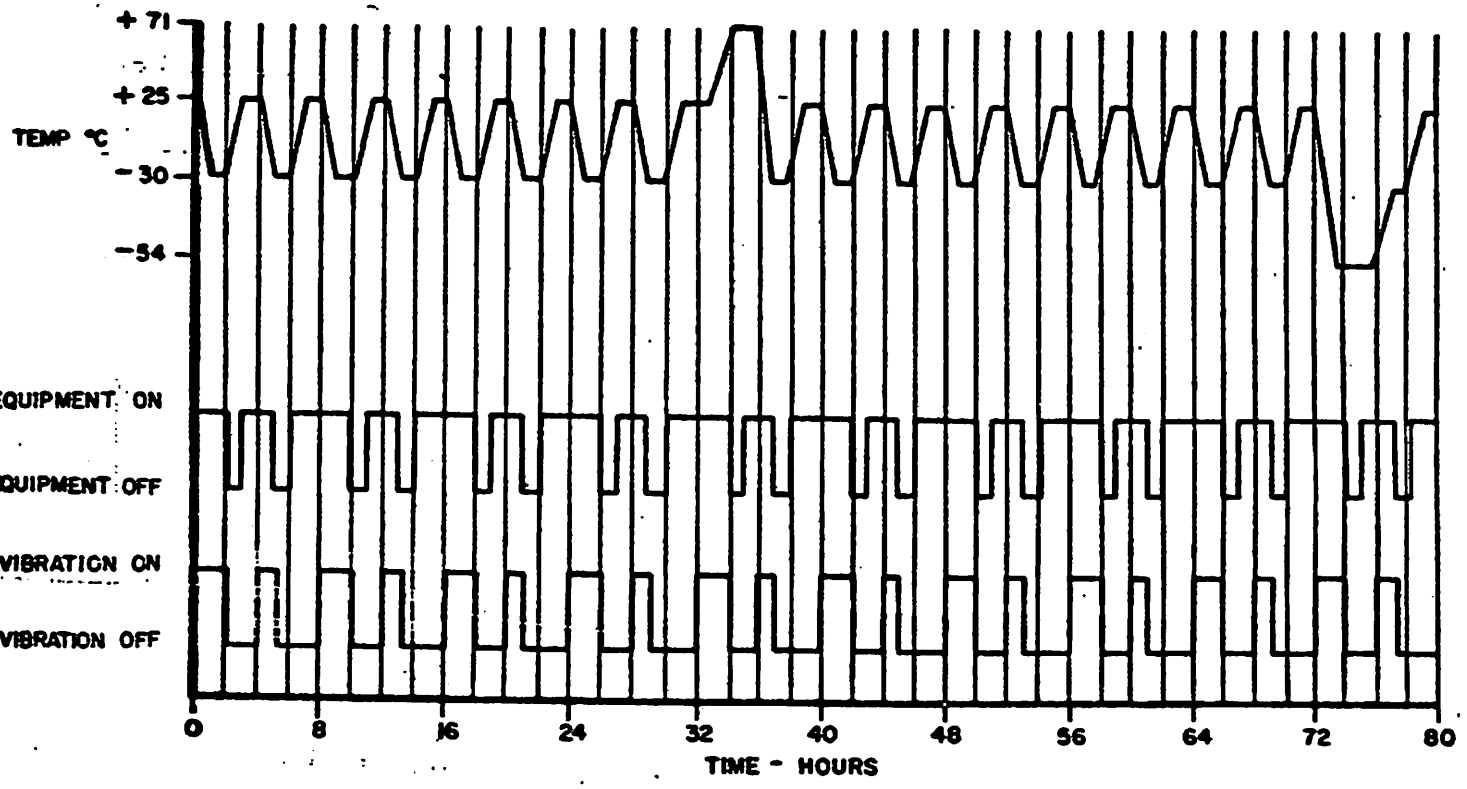
4.6.26 Acoustical noise. The transmitter shall be subjected to an acoustical noise test in accordance with MIL-STD-810, Method 515, procedure I, grade A. At the conclusion of this test, the transmitter shall be subjected to and shall pass the individual tests.

* 4.6.26.1 Output impedance. The transmitter shall be connected to a load of 1,000 ohms ±3 percent and rotated at 180 degrees per minute in both directions. The output shall be 1.35 ±.25 ma.

* 4.6.27 Weight and size. The transmitter shall be weighed and shall be less than 2.50 pounds. The case shall be measured and shall comply with figure 1.

4.6.28 Reliability. Prior to being subjected to the reliability test, the transmitters shall pass sampling plan A tests. After satisfactory completion of these tests, the transmitters, properly connected and with no power applied, shall be placed in test chambers which meet the environmental cycling conditions shown of figure 3. During the ON cycle, the transmitters shall be exercised by a 3-axis, 30° total movement, 10 cpm reversing scorsby action and by the vibration shown on figure 3. After each 160-hour cycle of operation, the transmitters shall meet the individual tests, except 4.6.7, 4.6.8, and 4.6.8.1. The tolerances shall be as specified in 4.6.28.1.1. Upon completion of this test, or upon reaching an accept decision, the transmitters shall meet the sampling plan A tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.



1) Temperature change >2°C (36°F)/minute average. Periods of +71°C (+160°F) & -54°C (-65°F), tests may be changed to coincide with working shift.
2) Equipment off period = 1/2 hour within the period shown.
3) Vibration on period = 3 hrs/8 hrs approx as shown.
4) Temp. chamber shall be stabilized at the temp. shown at each hour point
5) Environment cycle repeats at each 80 hours.
6) Vibration level = 0.2g 20 to 30 cps.
7) Monitoring of outputs for catastrophic failure each (on) cycle.
8) Monitoring of performance conducted each 160 hours.
9) Accumulated hours = all operating time..

FIGURE 3. Reliability Test Environment Cycles

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T11:10:07Z
Check the source to verify that this is the current version before use.

4.6.28.1 Longevity. At the completion of the test specified in 4.6.28, three of the transmitters shall continue to be tested for a total of 3,000 hours each.

* 4.6.28.1.1 Tolerances. The increase in tolerance for reliability and longevity testing shall be as follows:

Percent Increase in Tolerance	Reliability Test Time per Component	Tests
0	0-500 hours	4.6.2, 4.6.3, 4.6.4, 4.6.5, 4.6.6, 4.6.9, 4.6.10
0	500-1000 hours	4.6.4
5	500-1500 hours	4.6.2, 4.6.5
5	1000-1500 hours	4.6.4
10	1500-3000 hours	4.6.5
10	After 1500 hours	4.6.2, 4.6.4
15	After 3000 hours	4.6.5
25	500-1000 hours	4.6.3, 4.6.6, 4.6.9, 4.6.10
50	1000-1500 hours	4.6.3, 4.6.6, 4.6.9, 4.6.10
75	1500-3000 hours	4.6.3, 4.6.6, 4.6.9, 4.6.10
100	After 3000 hours	4.6.3, 4.6.6, 4.6.9, 4.6.10

* 4.6.28.1.2 After completion of the reliability test, when the transmitters have operated less than 3,000 hours, the following tolerances shall apply for sampling plan A tests:

a. Individual tests: As specified in 4.6.28.1.1

b. 4.6.7: No tolerance increase.

c. 4.6.8: Tolerances for 4.6.4 during voltage extremes shall be +12 and -20 percent of normal output at the respective rates-of-turn.

d. 4.6.8.1: Increases as specified for the individual tests for the appropriate number of operating hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

e. 4.6.11 and 4.6.12: Tolerance increases shall be as follows:

(1) 4.6.3: ±60μa
(2) 4.6.4: ±18 percent.
(3) 4.6.5: ±30 percent.

f. 4.6.13: Output signal shall not exceed ±60μa.

g. 4.6.14: Average value of signal integrated over at least 10 full cycles shall not exceed ±70μa.

h. 4.6.14.1: Increases as specified for the individual tests for the appropriate number of operating hours.

i. 4.6.15: No tolerance increase.

4.6.28.1.3 After 3,000 hours of operation, the following tolerances shall apply for sampling plan A tests:

a. Individual tests: As specified in 4.6.28.1.1.

b. 4.6.7: No tolerance increase.

c. 4.6.8: Tolerances for 4.6.4 during voltage extremes shall be +12 and -20 percent of normal output at the respective rates-of-turn.

d. 4.6.8.1: Increases as specified for the individual tests for the appropriate number of operating hours.

e. 4.6.11 and 4.6.12: Tolerance increases shall be as follows:

(1) 4.6.3: ±80μa
±20 percent
±35 percent.

f. 4.6.13: Output signal shall not exceed ±75μa.

g. 4.6.14: Average value of signal integrated over at least 10 full cycles shall not exceed ±80μa.

h. 4.6.14.1: Tolerances allowed for individual tests after 3,000 hours of operation.

i. 4.6.15: No tolerance increase.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:10Z
Check the source to verify that this is the current version before use.

4.6.28.2 Test information. Sixty days prior to starting this test, the contractor shall submit detailed test information to the procuring activity for approval. This information shall include but not necessarily be limited to:

a. Equipment used in monitoring performance

b. Periodicity of calibration checks to be made on monitoring equipment

c. Description of environmental test facilities to be used including recording facilities, methods of controlling heating and cooling facilities, and vibration equipment

d. Frequency of monitoring.

* 4.6.28.3 Test records. Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-STD-781: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.28.4 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.29 Maintainability demonstration plan. The contractor shall prepare a maintainability demonstration plan in accordance with MIL-STD-471 to determine that the requirements of 3.4.5 have been met.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, all levels. Preservation shall be in accordance with Method III of MIL-P-116, and packaging shall be in accordance with MIL-STD-794.

5.2 Packing

5.2.1 Level A. Transmitters preserved and packaged as specified in 5.1 shall be packed for export shipment in accordance with level A of MIL-STD-794.

5.2.2 Level B. Transmitters preserved and packaged as specified in 5.1 shall be packed in accordance with level B of MIL-STD-794.

5.2.3 Level C. Transmitters shall be packed in accordance with the best standard commercial practices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

5.3 Marking for shipment. Marking for shipment shall be in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

TRANSMITTER, RATE GYROSCOPE TRU-2A/A.

6. NOTES

* 6.1 Intended use. The TRU-2A/A rate-of-turn transmitter covered by this specification is intended for use in aircraft for sensing the rate at which the aircraft is turning and the relative direction of turn is equal to the rate-of-yaw. When mounted with the forward end elevated slightly, an anticipatory output is added as the aircraft is banked into the turn. At relatively low airspeeds, the rate-of-turn output is equal to rate-of-heading change so that time to turn a given arc is known. At high airspeeds or extreme maneuvers, it is necessary to take into account the difference between the rate-of-yaw output and rate-of-heading change.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. When sampling plan C tests will not be conducted

d. Levels of packaging and packing desired

e. All failures occurring during qualification of quality conformance testing. This report will be submitted within 2 working days.

6.3 Definitions

6.3.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.2 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

* 6.3.3 Discrimination ratio. The discrimination ratio is the ratio of the contractually specified MTBF to the minimum acceptable MTBF. In the event that any conflict in definition exists between this specification and any other document, the definition of this specification will govern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.

* 6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Directorate of Airframe Engineering, Attn: ENFIF, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

6.5 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 71, 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:10Z
Check the source to verify that this is the current version before use.